Truncation Impact Score: A Diagnostic Metric Framework for Quantifying Salient Content Loss in Long-Form Biomedical Summarization

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ABSTRACT

Summarizing long biomedical articles is a difficult task for AI models because most of them can only read a limited portion of a document at a time. When an article is too long, the model simply cuts off the rest — which means important information from the middle and later sections of the article never gets used. This study introduces the **Truncation Impact Score (TIS)**, a new tool that measures how much valuable content a model misses before it even starts generating a summary. Unlike existing evaluation tools that only look at the quality of the output, TIS works beforehand — identifying how much is lost before any summary is made. To test this, three AI summarization models were trained and compared using **12,000 biomedical articles from PubMed**. Results showed that the standard model **(BART) still performed best overall**, but TIS revealed that as more content gets cut off, the performance gap between models gradually closes. This means TIS can be used as a practical guide to help researchers decide which model is best suited for a given document — without needing to run expensive computations first.

INTRODUCTION

The exponential growth of biomedical literature presents a formidable barrier to evidence synthesis. The PubMed repository currently indexes over 40 million citations [38], and the volume of new publications continues to rise. Automatic abstractive summarization offers a scalable mechanism to assist researchers and clinicians in navigating this corpus. However, the dominant architectures for sequence-to-sequence generation—most notably BART [23]—operate under a severe architectural constraint: a 1,024-token input limit. Given that the average PubMed article in our corpus spans 2,729 tokens (train split), standard models silently discard approximately 62% of source content before a single output token is generated.

This truncation is not uniform in its consequences. Biomedical articles follow the IMRAD structure (Introduction, Methods, Results, Discussion), where the methodology and empirical findings—the most scientifically critical content for a summary—appear in the middle and later sections of the document, well beyond the 1,024-token boundary for most papers. The resulting information loss is systematic and domain-specific, yet it is invisible to output-centric evaluation metrics such as ROUGE [24] and BERTScore [56], which assess the quality of what was generated rather than what was discarded.

Long-context architectures such as the Longformer Encoder-Decoder (LED) [3] provide an archi-tectural remedy by extending the context window to 16,384 tokens via sparse attention. However, the field lacks a diagnostic mechanism to predict when this additional capacity is necessary and to what degree it compensates for the truncation bottleneck. This gap motivates the Truncation Impact Score (TIS), a pre-inference, model-agnostic metric that quantifies salience-weighted content loss before any model inference is performed.

The input to our pipeline is a set of PubMed biomedical article-abstract pairs. We compute TIS for each article as a diagnostic score, then train and evaluate three summarization systems using LoRA fine-tuning: a BART-large baseline, an LED-large long-context model, and a BART-Oracle intelligent truncation baseline.

RESEARCH QUESTIONS

1. What fraction of **summary-salient content** in PubMed biomedical articles falls beyond a 1,024-token context window, and how does this vary across documents?
2. Can TIS reliably predict and explain performance differences between **truncation-limited models (BART) and long-context models (LED)** across documents with varying truncation severity?

RECOMMENDATIONS

- **Explore TIS in unsupervised settings** (no reference summaries) using TextRank or lead-extractive salience approximations
- **Extend the study to domains without positional lead bias** — such as legal contracts or clinical discharge summaries — where LED's architectural advantage may manifest in absolute, not only relative, terms
- **Evaluate beyond ROUGE using faithfulness metrics** (SummaC, FactCC) to capture the full truncation-quality relationship
- **Use TIS as a computational routing signal in production pipelines:** compute TIS cheaply to decide when expensive LED inference is justified, reducing inference cost and carbon footprint

REFERENCES

For References & **FULL COPY** of the paper, kindly **SCAN** the **QR Code**.

METHODOLOGY

Dataset. PubMed Article Summarization Dataset: 12,000 training pairs, 500 validation, 1,500 test. Average article length: 2,729 tokens (train); average abstract: ~182 tokens.

TIS Formula. TIS is computed as the salience-weighted fraction of reference-relevant content beyond context limit C:

$$\mathrm{TIS}(d, C, r) = \Sigma\, w(si)\cdot\mathrm{overlap}(si, r) \text{ [for si beyond C]} \;/\; \Sigma\, w(sj)\cdot\mathrm{overlap}(sj, r) \text{ [for all sj]}$$

Where w(si) is the global TF-IDF importance weight fitted on 5,000 training articles, and overlap(si, r) is unigram recall against the reference abstract. TIS = 0 means all salient content is within the window; TIS = 1 means all salient content has been discarded.

TIS Difficulty Bins

Bin	TIS Range	Records	Share	Mean TIS
Low	< 0.2	349	23.3%	0.0353
Medium	$[0.2, 0.5)$	303	20.2%	0.3581
High	≥ 0.5	848	56.5%	0.6966

Three Model Systems Compared

- **BART-large** (1k Baseline) — Standard right-side truncation to 1,024 tokens
- **LED-large** (4k Long-Context) — Sparse attention, up to 4,096 tokens; initialized from BART-large weights for a controlled comparison
- **BART-Oracle** (1k Smart-Select) — TF-IDF-ranked sentence selection within 1,024 tokens; isolates the effect of smarter truncation vs. architectural extension

LoRA Fine-Tuning. Both BART and LED were fine-tuned using LoRA (rank 8 and rank 4 respectively) on a single NVIDIA A5000 24GB GPU, AdamW optimizer, learning rate 3×10^{-5}, 3 epochs.

EXPERIMENTAL RESULTS

Biomedical articles from PubMed were collected and used to test the study. As shown in Figure 1, on average each article contains around 2,700 tokens — far more than what BART can process at one time (only 1,024 tokens). This means that before BART even begins generating a summary, it has already discarded the majority of the article's content. Figure 2 further illustrates this problem, showing that more than half of the test articles lost over 50% of their important content due to this limitation, confirming that severe truncation is a common problem, not a rare exception.

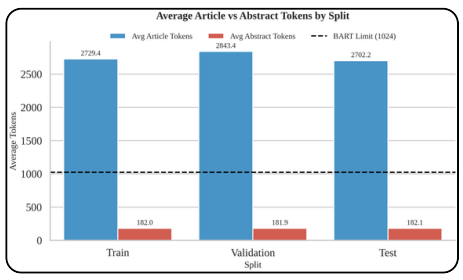


Figure 1: Dataset Composition and Token-Length Profiles

Three models were trained and tested to compare their summarization performance. As shown in Figure 3, BART-large produced the best overall results with the highest scores across all evaluation metrics. Surprisingly, LED-large — which can process up to 4,096 tokens at a time — performed lower than BART. This unexpected result is explained by the way PubMed abstracts are written: they tend to closely reflect the introduction and background sections of an article, which happen to fall within BART's token limit. A third model, BART-Oracle, was also tested using a smarter way of choosing which sentences to feed into BART. While it recovered more relevant content from throughout the article, Figure 3 also shows that its summaries were less fluent and coherent compared to the standard BART approach.

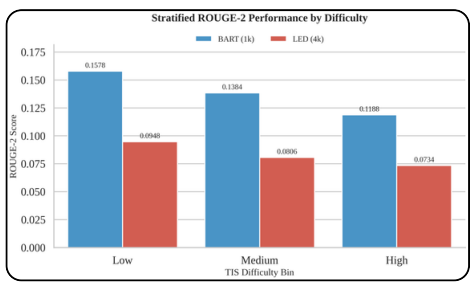


Figure 2: Stratified ROUGE-2 Performance by Difficulty

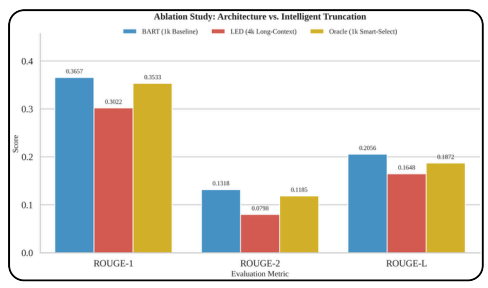


Figure 3: Ablation Study: Architecture vs. Intelligent Truncation

The most important finding of this study comes from grouping the test articles by their TIS score. As shown in Figure 2, as the TIS score increased — meaning more important content was being cut off — BART's performance dropped steadily and predictably. At the same time, Figure 2 shows that the performance gap between BART and LED gradually narrowed across the Low, Medium, and High TIS groups. This pattern confirms that TIS works as a reliable early warning signal: the higher the TIS score of a document, the more it would benefit from being processed by a long-context model like LED. In practical terms, this means researchers and developers can use TIS to decide which model to use for a given document before running any costly computation, saving both time and resources.